PROJECT DIRECTIVE & PROBLEM STATEMENT

FROM:	VP of Business Analytics & Strategy	**ROLE:**	Lead Data Analyst
PROJECT:	E-Commerce Performance & Governance	**DATASET:**	Relational Extract (1,55,847 Rows)

1. EXECUTIVE BACKGROUND & BUSINESS MANDATE

We have consolidated sales transactions, customer demographics, product catalog listings, and return logistics from multiple database systems. The raw dataset contains over **1.55 lakh order records**. Due to disparate logging mechanisms across payment gateways and shipping partners, the data contains significant noise, including case mismatches, mixed date formats, and duplicate records.

2. DATASET SCOPE & ARCHITECTURE BASELINE

1,55,847
ORDERS TABLE
Sales, Dates, Payments, Ships

15,300
CUSTOMERS TABLE
Profiles, States, Signups

192
PRODUCTS TABLE
Catalog, Pricing, Category

8,542
RETURNS TABLE
Reasons & Refund Status

3. DATA GOVERNANCE & HYGIENE MANDATES

Strict Data Integrity Guidelines (Zero Dirt Rule)

Before presenting executive metrics or building dashboards, you must execute an ETL pipeline that enforces the following requirements:

- **Strict Identity Integrity:** Primary identifiers (`customer_id`, `order_id`, `sales_amount`) contain 0% missing values. You must **NEVER** allow generic fallback values such as *"Unknown"*, *"N/A"*, or *"Null"* to appear in top executive views (e.g., Top 3 Customer Leaderboards).
- **Text & Categorical Standardization:** Normalize casing inconsistencies across categorical variables (`payment_method`, `ship_mode`, `category`, `state`). Values like `UPI`, `upi`, and `Credit Card`, `credit card` must be grouped into single standardized entities.
- **Date Uniformity:** Parse all polymorphic date entries (`YYYY-MM-DD`, `DD/MM/YYYY`, `MM-DD-YYYY`) into a single uniform ISO standard date format (`YYYY-MM-DD`).
- **Deduplication Protocol:** Identify and eliminate transaction duplicates across the 1.55L dataset while maintaining cross-table relational integrity.

4. DATA ISSUES & CLEANING ACTION PLAN

Target Field	Raw Condition / Noise	Required Action	Dashboard Status
Customer Identifiers	Fully intact (0% Nulls)	Preserve & Validate	Clean Visible
Payment & Ship Methods	Mixed Casing (`upi`, `UPI`, `COD`, `cod`)	UPPERCASE / TitleCase	Requires ETL
Order Dates	Inconsistent Formats (`DD/MM/YYYY`, etc.)	Parse to YYYY-MM-DD	Requires ETL
Transaction Rows	~1,200 Duplicate Records	Deduplicate on Order ID	Requires ETL

5. STRATEGIC BUSINESS DELIVERABLES & ANALYTICS FOCUS

Once data governance standards are established, you are expected to analyze the dataset and deliver answers to the following operational questions for the upcoming leadership board meeting:

A. Financial & Sales Revenue Analysis

- **Core Revenue Metrics:** Calculate Total Gross Sales, Net Sales (accounting for returns and discounts), Total Order Volume, and overall Average Order Value (AOV).
- **Discount Optimization:** Determine total margin spend across discounts and analyze whether discounting correlates directly with order volume or net revenue loss.
- **Customer Value Benchmarking:** Extract the **Top 3 Customers by Total Net Spend**. Provide a clean summary containing Customer Name, Total Orders Placed, State, and Lifetime Net Spend. Ensure no missing or placeholder names exist in this view.

B. Product Category & Return Mitigation

- **Category & Margin Drivers:** Identify top-performing product categories by net revenue, along with the Top 5 most profitable products versus low-margin items.
- **Return Rate Breakdown:** Compute overall Return Rate % (`Total Returned Items / Total Sold Items`). Pinpoint which specific Category and Sub-Category suffer from highest return rates.
- **Root Cause Analysis:** Segment returns by **Return Reason** (e.g., Defective, Size Issue, Wrong Item) and evaluate financial impact by processing status (`Processed` vs `Pending`).

C. Logistics & Customer Behavior Insights

- **Channel Preference:** Analyze customer preferences across Payment Methods and Shipping Modes by region/state.
- **Fulfillment Speed vs Return Correlation:** Evaluate if expedited shipping options (`Same Day` / `First Class`) show significantly higher return rates compared to standard shipping.
- **Customer Retention:** Calculate Repeat Purchase Rate to evaluate customer loyalty across regions.

6. REQUIRED FINAL DELIVERABLES & TIMELINE

Deliverable	Description / Content Expectations	Format Required
1. ETL & Data Pipeline	Cleaned CSVs, Reconciliation log (Raw vs Cleaned row counts), Deduplication report.	Cleaned CSV / SQL Scripts
2. Executive Dashboard	Interactive views with KPI cards (Revenue, AOV, Return Rate), Sales Trends, Top 3 Customer Card.	BI Dashboard / Wireframe
3. Strategic Action Plan	3-4 concrete business recommendations on return reduction, product margin fixes, and discount rules.	Executive Summary Deck